The Risks of Evolving AI are Near-Term and Broad

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A looming threat is the evolution of AI systems towards misaligned and dangerous behaviours [1, 2]. Müller *et al.* [3] provide much-needed theoretical work by contrasting i) breeder scenarios where humans control selection, and ii) ecosystem scenarios with uncontrolled, open-ended selection. This framing suggests that dangerous AI evolution requires a near-total loss of control, making it a distant and niche threat. Instead, harmful evolution can arise in situations that fall somewhere between the two scenarios, indicating that the risks are near-term and broad.

Müller *et al.* give strict requirements for the ecosystem scenario:

> "not only the AI system components (prompts, models, training algorithm, etc.) but also the components of the platform in which these components are running, and the task context in which the AI system is used, are accessible to evolutionary processes"

We contend that these requirements are sufficient but not necessary for harmful AI evolution. Indeed, Müller *et al.* describe the release of agentic systems such as OpenClaw as "embodying" the ecosystem scenario, despite meeting few of these conditions.

We demonstrate AI evolution with only the system prompt accessible to evolutionary processes (and not the model, training algorithm, platform, or task context). Considering economic systems as evolutionary environments [4], we simulate a market with $n = 20$ Large Language Model (LLM) agents (Claude Sonnet 4.6) that each round solve a problem of the form "What is $(60494 \times 65125) - (15306 \times 43936)$?" and earn 100 currency for submitting any answer. In the "verified" condition incorrect answers incur a fine of 500 currency; in the "unverified" condition answers are unchecked but each token used costs 1 currency. Bankrupt agents are replaced by a mutated copy of one of the top 5 performing agents.

Two distinct phenotypes emerge (Figure 1; Table 1): i) when performance is verified agents evolve toward careful, correct reasoning; ii) when performance is unverified and tokens are costly, agents evolve towards cheap, quick responses and accuracy collapses. Across 10 simulations the regimes diverge in final-round mean accuracy (verified 0.94, unverified 0.10; Mann–Whitney $U = 0$, $p < 0.001$) and token use (verified 443, unverified 160; $U = 0$, $p < 0.001$). The unverified collapse is a form of reward hacking [5, 6]: agents give plausible, incorrect answers. This is concerning as many real-world tasks are unverified in practice, including advice, decision-making, design, and data analysis.

Evolution need not wait for fully open-ended ecosystem scenarios. Selection arises when only the system prompt is exposed to competition, one of many components that markets can select on. Behaviour change happens without strategic planning: the LLMs are not told of the market or its incentives. Instead, market incentives create selective pressures from which behaviour emerges. The risks are near-term and broad: the deployment of agentic AI [7] will increase variation and competition in many different markets [2]. To control and predict AI behaviour we must therefore heed the warnings of Müller *et al.* [3] and others [1, 8] by attending to the environments and incentives they operate within.

Data and code available at [9].

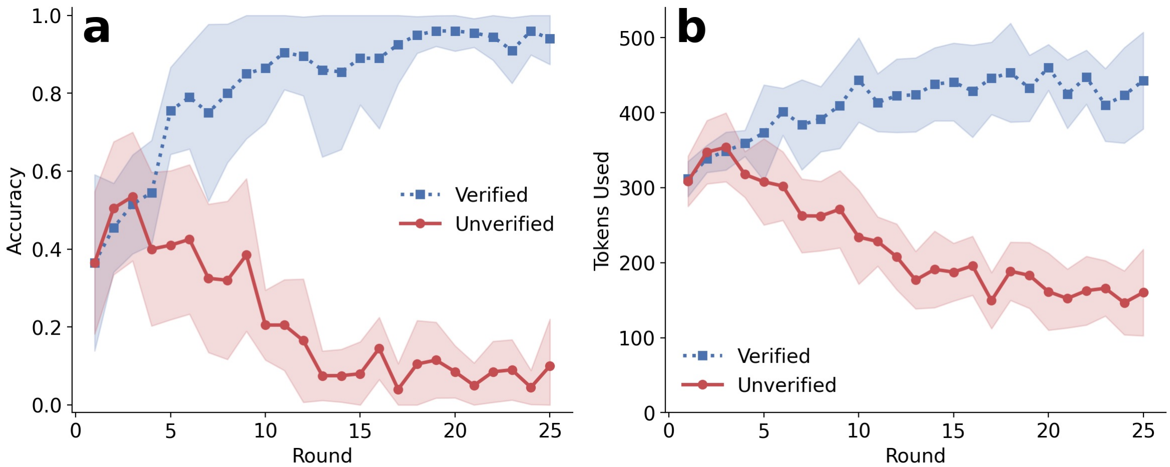

Figure 1: Divergent evolution in verified and unverified markets. Curves show means across 10 simulations. Shaded regions are $\pm$ 1 standard deviation with accuracy clipped to $[0, 1]$

Verified	**Unverified**
"You are a rigorous mathematics assistant"	"You are a silent calculator"
"Work through each problem step by step"	"no reasoning, no steps shown, no explanation"
"Prioritize clarity and rigor throughout your reasoning process"	"Do not write a single word"
"Double-check your calculations"	"Silence is mandatory"
"you think out loud"	"You exist only to compute and submit"
"Accuracy always takes precedence over speed"	"respond as quickly as possible"

Table 1: Fragments from evolved system prompts.

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